A Positive Spectral Discrepancy Criterion Equivalent to the Riemann Hypothesis under a Distributional Weil Framework

Abstract

We construct a test function H_0 in the admissible Weil test space and define a discrepancy functional $Q = -\langle S, H_0 \rangle$ associated with the distribution $S = \partial_t^2 \log |\xi(\frac{1}{2} + it)|$. We establish a criterion equivalent to the Riemann Hypothesis under the stated analytic framework: $Q = Q'_{\mathrm{RH}}$ holds if and only if all nontrivial zeros satisfy $\operatorname{Re} \rho = \frac{1}{2}$.

1 Introduction

1.1 Objective

The goal of this work is not to prove the Riemann Hypothesis but to construct a functional

$$Q - Q'_{\mathrm{RH}} = \sum_{\rho} \Delta_H(\gamma_\rho, \delta_\rho) \tag{1}$$

and to establish the equivalence

$$Q = Q'_{\mathrm{RH}} \iff \delta_\rho = 0 \text{ for all } \rho. \tag{2}$$

1.2 Relation to Weil's explicit formula

We use Weil's explicit formula in its tempered-distribution form $\langle S, H \rangle = M(H) + P(H) + Z(H)$ and the extension due to Barner admitting test functions beyond the Schwartz class. The new content of the present work is the positivity structure of the discrepancy Δ_H on the nontrivial zero spectrum.

1.3 Contributions

- **Kernel construction:** the single-zero kernel K_δ with moment cancellation $M_0 = M_1 = 0$;
- **Absolute interchange:** the zero contribution $\sum_\rho |\langle K_\rho, H_0 \rangle| < \infty$;
- **Positive discrepancy identity:** $Q - Q'_{\mathrm{RH}} = \sum_\rho \Delta_H(\gamma_\rho, \delta_\rho)$ with $\Delta_H(\gamma_\rho, \delta_\rho) > 0$ for $\delta_\rho \neq 0$;
- **Rigidity criterion:** $Q = Q'_{\mathrm{RH}}$ if and only if all zeros lie on the critical line.

2 Definitions

Throughout, $\rho = \beta + i\gamma$ denotes a nontrivial zero of ξ, and

$$\delta_\rho = \beta - \tfrac{1}{2}, \qquad |\delta_\rho| < \tfrac{1}{2} \tag{3}$$

by the critical strip. The discrepancy functional is

$$Q = -\langle S, H_0 \rangle, \qquad S = \partial_t^2 \log |\xi(\tfrac{1}{2} + it)|, \tag{4}$$

and the reference functional is

$$Q'_{\mathrm{RH}} = \sum_\rho w_H(\gamma_\rho, 0), \tag{5}$$

obtained via the projection $\Pi(\rho) = \frac{1}{2} + i\gamma_\rho$ (preserving ordinate and multiplicity); it is a reference spectrum, not an assertion that $\Pi(\rho)$ is a zero of ζ. obtained by projecting each zero onto the critical line while preserving its ordinate and multiplicity; no RH assumption enters its definition.

3 Kernel Lemma

Proposition 3.1 (Single-zero kernel)**.** *The single-zero kernel $K_\delta(x) = 2(x^2 - \delta^2)/(x^2 + \delta^2)^2$ satisfies*

$$M_0 = \int K_\delta = 0, \qquad M_1 = \int x K_\delta = 0, \qquad |M_2| = \left| \int x^2 K_\delta \right| \le C|\delta|. \tag{6}$$

Proof. $M_0 = 0$ by the identity $\int \log(x^2 + a^2)\, dx = 2\pi a$ in the regularized sense; $M_1 = 0$ by parity; $M_2 = \delta \int y^2 Q_{\eta/\delta}(y)\, dy$ with the scaled profile Q_λ, and the scale-invariant bound $\sup_{\lambda < 1/2} \int y^2 |Q_\lambda(y)|\, dy < \infty$ yields $|M_2| \le C|\delta|$. ☐

Proposition 3.2 (Absolute interchange)**.**

$$\sum_\rho |\langle K_\rho, H_0 \rangle| < \infty, \tag{7}$$

and consequently $\langle \sum_\rho K_\rho, H_0 \rangle = \sum_\rho \langle K_\rho, H_0 \rangle$.

Proof. By the moment cancellation of Proposition 3.1, $|\langle K_\rho, H_0 \rangle| \ll |\delta_\rho|\, |H_0''(\gamma_\rho)|$. Since $|\delta_\rho| < \frac{1}{2}$ (critical strip, not RH) and $H_0''(t) = O(t^{-2})$ with $N(T) = O(T \log T)$, the series $\sum_\rho |\delta_\rho H_0''(\gamma_\rho)| < \infty$. ☐

4 Distributional Weil Framework

Pairing definition. The distribution $S = \partial_t^2 \log |\xi(\frac{1}{2} + it)| \in \mathcal{S}'$ is paired with H_0 by integration by parts:

$$\langle S, H_0 \rangle = \langle \log |\xi(\tfrac{1}{2} + it)|, H_0'' \rangle. \tag{8}$$

This avoids requiring $H_0 \in L^1$: although $H_0(t) \sim \log |t|$, we have $H_0''(t) = O(t^{-2})$, hence $H_0'' \in L^1$ and the pairing is finite. Note that H_0 is even, real, C^∞, and tempered.

Weil formula scope. We use the distributional extension framework of Weil's explicit formula and verify the required pairings explicitly for H_0: the zero part is defined through the products $\widehat{K}_\delta\, \widehat{H}_0 = \frac{\pi}{2}(1 + |u|)e^{-(1+|\delta|)|u|} \in L^1$, and the archimedean/prime parts are computed in the tempered distribution framework of Schwartz–Hörmander. The present work does not reprove the explicit formula; it verifies that all pairings are well-defined for H_0.

Proposition 4.1 (Distributional Weil compatibility). *With the pairing* (8)*, the distributional form of Weil's explicit formula applies:* $\langle S, H_0\rangle = M(H_0) + P(H_0) + Z(H_0)$*, where* $M(H_0)$ *collects the Gamma and archimedean terms,* $P(H_0)$ *the prime terms, and* $Z(H_0)$ *the zero contribution* $\sum_\rho \langle K_\rho, H_0\rangle$ *(defined through the products* $\widehat{K}_\delta \widehat{H}_0 \in L^1$*; all zeros counted with multiplicity).*

5 Positive Spectral Discrepancy

Lemma 5.1 (Positivity rigidity on the nontrivial zero spectrum). *For every nontrivial zero ordinate* $|\gamma_\rho| \geq \gamma_1$ *with* $\gamma_1 = 14.1347\ldots$*, one has*

$$\Delta_H(\gamma_\rho, \delta_\rho) > 0 \quad \textit{whenever } \delta_\rho \neq 0, \tag{9}$$

and $\Delta_H(\gamma_\rho, 0) = 0$.

Proof. By the closed form

$$\Delta_H(\gamma, \delta) = |\delta|\, \widetilde{c}_H(\gamma, \delta), \qquad \widetilde{c}_H(\gamma, \delta) > 0 \quad (|\gamma| \geq \gamma_1), \tag{10}$$

where $\widetilde{c}_H = \Delta_H/|\delta|$ and positivity follows from the numerator $N(x) = (1/2)(a^2(a+1) + x\gamma^2)(1+\gamma^2)^2 - (a^2+\gamma^2)^2$, $a = 1 + x = 1 + |\delta|$: $N(0) = 0$ and $\partial N/\partial x \geq \gamma^6/2 + \gamma^4 + \gamma^2/2 - 6\gamma^2 - 13.5 > 0$ for $\gamma \geq 2$, so $N(x) > 0$ for $x > 0$. The positivity is pointwise on the spectral set; no uniform lower bound is needed. Moreover $\widetilde{c}_H(\gamma, \delta) \sim 1/(2\gamma^2)$ as $\gamma \to \infty$, and $\sum_\rho \Delta_H(\gamma_\rho, \delta_\rho) < \infty$ since $\Delta_H \sim |\delta_\rho|/(2\gamma_\rho^2)$ and $\sum_\rho \gamma_\rho^{-2} < \infty$. □

6 Main Theorem

Theorem 6.1. *Let* $Q = -\langle S, H_0\rangle$ *and* $Q'_{\mathrm{RH}} = \sum_\rho w_H(\gamma_\rho, 0)$*. Then*

$$Q - Q'_{\mathrm{RH}} = \sum_\rho \Delta_H(\gamma_\rho, \delta_\rho), \tag{11}$$

with $\Delta_H(\gamma, \delta) = w_H(\gamma, \delta) - w_H(\gamma, 0) \geq 0$ *and equality if and only if* $\delta = 0$*. Hence*

$$Q = Q'_{\mathrm{RH}} \iff \forall \rho,\ \delta_\rho = 0 \iff \forall \rho,\ \operatorname{Re} \rho = \tfrac{1}{2}. \tag{12}$$

Proof. The identity follows from Proposition 3.2 (interchange) and Proposition 4.1 (Weil interface), the positivity from Lemma 5.1, and the equality condition from the positive-term argument: $\sum_\rho \Delta_\rho = 0$ with $\Delta_\rho \geq 0$ forces $\Delta_\rho = 0$ for every ρ, hence $\delta_\rho = 0$ for every ρ by Lemma 5.1. □

References

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[3] L. Schwartz, *Théorie des distributions*, Hermann, Paris, 1950–1951.

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A Kernel estimates

The scaled profile $Q_\lambda(y)$ satisfies the far-field estimate $Q_\lambda(y) = -(\lambda^2/2)/(y^2+1)^2 + R_\lambda(y)$ with $|R_\lambda(y)| \le C/(y^2+1)^3$, giving the moment bound $|M_2| \le C|\delta|$ and the asymptotic $\widetilde{c}_H(\gamma,\delta) \sim 1/(2\gamma^2)$.

B Distributional pairings

$S = \partial_t^2 \log|\xi(\frac{1}{2}+it)| \in \mathcal{S}'$ via the pairing $\langle S, \phi\rangle = \langle \log|\xi(\frac{1}{2}+it)|, \phi''\rangle$. At a zero, $\partial_t^2 \log|t-\gamma_\rho| = -\operatorname{Pf}\frac{1}{(t-\gamma_\rho)^2}$ (Hadamard finite part, not δ'), and the single-zero kernel is $K_\rho = -\operatorname{Pf}\frac{1}{(t-\gamma_\rho)^2} +$ regular correction, consistent with K_δ; multiplicity m_ρ is carried by the kernel, conjugate zeros are summed with the same convention, and all regular terms are absorbed into $M(H_0)$.

C Conventions

Fourier transform: $\widehat{H}(u) = \int_{\mathbb{R}} H(t) e^{-2\pi i u t}\, dt$ (all formulas follow this normalization). Xi normalization: $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Zero counting: $\rho = \frac{1}{2} + \delta_\rho + i\gamma_\rho$ with multiplicity; sums over all nontrivial zeros. Prime-term sign is fixed by the Fourier convention.

Limitations

The validity of the criterion depends on the stated analytic framework and the applicability of the invoked explicit formula. The equivalence theorem is conditional on the stated distributional framework and on the validity of the explicitly verified analytic identities. Independent verification of the key lemmas by the community is required before any stronger claim is made.